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RESEARCH ARTICLE

Life Cycle Environmental Impact Analysis of Lithium Carbonate Production from Lepidolite and Comparison with Lithium Extraction from Spodumene

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Abstract

With the rapid growth in energy storage demand, lepidolite is emerging as the third major source of lithium extraction after brine and spodumene. However, the environmental impacts of lithium extraction from lepidolite remain unclear. This study employs a life cycle assessment (LCA) approach to establish a cradle-to-gate evaluation framework covering the entire process from lepidolite ore mining to lithium carbonate production. The results indicate that the environmental impact of lithium extraction from lepidolite is significantly higher than that from spodumene. Specifically, the endpoint environmental impacts in the categories of human health, ecosystem quality, and resources are 6.23 times, 7.84 times, and 21.61 times greater, respectively. The beneficiation stage accounts for 76.83% of impacts on human health and 81.06% on ecosystems, while the lithium carbonate production stage contributes 86.64% to resource consumption. Key factors influencing environmental impact include electricity, diesel, sulfuric acid, and other raw materials and energy sources. Sensitivity analysis shows that lithium extraction from lepidolite is highly sensitive to the lithium oxide content in the ore and the electricity mix. Increasing the lithium oxide content in lepidolite ore to 0.8% can reduce the overall environmental impact of Land use by up to 34%. Furthermore, as the electricity structure becomes greener, the global warming potential could be reduced to 21% of the current level. Techno-economic analysis indicates that increasing the market price of lithium carbonate and adopting resource recovery measures can significantly improve the economic viability of lithium extraction from lepidolite. This study provides a new perspective for understanding the environmental impacts and economic benefits of lithium extraction from lepidolite.

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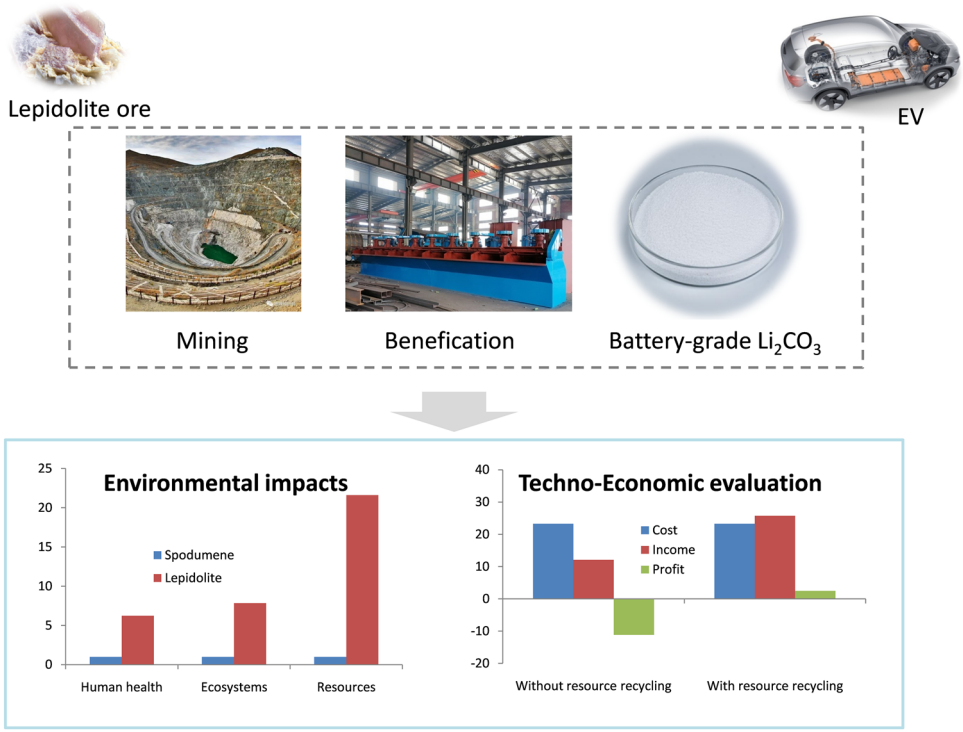
Graphical Abstract



Keywords Lepidolite · Lithium carbonate · Life cycle assessment · Techno-economic evaluation

Abbreviations

CCPG	Central china power grid
EVs	Electric vehicles
GHG	Greenhouse gas
GWP	Global warming potential
H_2SO_4	Sulfuric acid
ICE	Internal combustion engine
LCA	Life cycle assessment
LCI	Life cycle inventory
LCIA	Life cycle impact assessment
LIBs	Lithium-ion batteries
Li	Lithium
Li_2CO_3	Lithium carbonate
Li_2O	Lithium oxide
Na_2CO_3	Sodium carbonate
NaOH	Sodium hydroxide
Na_2SO_4	Sodium sulfate
NCM	Nickel–cobalt–manganese oxide
PAM	Polyacrylamide
TCM	Techno-economic cost model

Introduction

Electrification of transportation is an effective way to address air pollution and climate change [1–3]. It has been reported that electric vehicles (EVs) reduce greenhouse gas emissions by 18.3% to 22.6% over their entire life cycle compared to internal combustion engine (ICE) vehicles [4–6], and can significantly improve urban air quality [3, 7]. The majority of electric vehicles (EVs) rely on lithium-ion batteries (LIBs) as their driving force [8, 9]. The rapid development of the EV market has created a huge demand for power lithium-ion batteries, with projections indicating that this demand will reach 2,500 GWh by 2030 [10]. The use of critical metals such as lithium, cobalt, and nickel in LIBs, combined with pollution generated during their production, has raised concerns regarding resource availability and environmental impacts [11–17].

Lithium is primarily found in nature in seawater, brines, and ores. Seawater contains a significant amount of lithium, but its concentration is very low, at only about 0.17 ppm. This low concentration makes it difficult to develop effective extraction methods. Currently, the technology for extracting lithium from seawater is not yet mature [18]. More than

two-thirds of commercially exploitable lithium resources are found in brines, which are mainly distributed in South American countries such as Argentina, Bolivia, and Chile. These brines have a Lithium content of about 0.15% and are currently the main source of high-quality and low-cost lithium supply [19]. However, the enrichment process of brines typically relies on solar evaporation, which results in a long production cycle and makes it difficult to rapidly expand production capacity. The primary commercially viable lithium minerals are spodumene and lepidolite. High-grade spodumene (with Li_2O content generally above 1.0 wt.%) is predominantly found in Australia [20]. Lepidolite is primarily located in Yichun City, Jiangxi Province, China, and is characterized by its low lithium content, with an average Li_2O grade of approximately 0.4 wt.% [21, 22]. Currently, the Li_2CO_3 on the market mainly extracted from brines and spodumene [23]. The lithium production and resource reserves of major countries in 2024 are presented in Table 1 [24]. However, given the massive projected demand for lithium-ion batteries, lepidolite is expected to become the third most significant source of lithium extraction.

Given its role as a critical raw material for LIBs, the environmental impact of lithium extraction has consistently been a major concern within the industry. In a "cradle-to-grave" life cycle assessment, Notter DA et al. [8] analyzed EVs using lithium sourced from the Atacama Salt Lake brine in Chile. Their results showed that the battery accounted for 15% of the total lifecycle environmental impact of the EVs, with lithium extraction itself contributing less than 2.3%. Thus, they concluded that the impact of brine-based lithium extraction on the entire lifecycle of an EV is negligible. However, Schenker V et al. [25] analyzed lithium extraction processes from brines at five different locations with varying compositions and techniques, finding that some of these processes have a significant environmental impact over the entire life cycle of LIBs. Jiang S et al. [26] analyzed the environmental impacts of lithium extraction from spodumene and compared it with the indicators of lithium extraction from brine. They found that all impact categories of the spodumene route were much greater than those of the brine route, with differences as high as 60.4 times. The use of LIBs packs based on spodumene extraction increased acidification and global warming potential by 17 to 32% compared to those based on brine extraction. The study by Kelly JC et al. [27] revealed that differing lithium sources caused variations of up to 20% and 45% in greenhouse gas (GHG) emissions for NMC811 and NMC622 cathodes, respectively. At the full battery level, these variations were lower but still significant, reaching up to 9% for NMC811 and 20% for NMC622. Theses study further proves that life cycle assessments of lithium-ion batteries and EVs must fully consider the differences in lithium sources.

The extraction of lithium from lepidolite differs significantly from that of spodumene [28, 29]. The dominant process for spodumene is the sulfuric acid method, whereas the sulfate method is typically employed for lepidolite. These differences in Li_2O content and process chemistry consequently lead to distinct environmental impact profiles for the two ore types. However, to the best of our knowledge, few studies have quantified the life cycle environmental impacts of lepidolite-sourced lithium. Given this gap, the question of whether lepidolite introduces additional environmental burdens has attracted significant attention from both academia and industry.

To bridge this gap, further research is required to comprehensively assess the environmental impacts of lithium extraction from lepidolite and compare these impacts with those associated with other lithium sources. This study focuses on lithium extraction from lepidolite, establishing a cradle-to-gate life cycle assessment model that encompasses three key production stages: lepidolite ore mining, beneficiation processing, and lithium carbonate production. The environmental impacts across the entire life cycle are quantitatively analyzed and compared with those of the spodumene-based lithium extraction pathway. The findings of this study can help identify the environmental hotspots associated with lepidolite-based lithium extraction and contribute to the sustainable development of the lithium battery industry.

Table 1 World lithium production and reserves in 2024 (metric tons)

	Mine production		Reserves
	2023	2024	
Argentina	8,630	18,000	4,000,000[a]
Australia	91,700	88,000	57,000,000[b]
Brazil	5,260	10,000	390,000[b]
Canada	3,240	4,300	1,200,000[a, b]
Chile	41,400	49,000	9,300,000[a]
China	35,700	41,000	3,000,000[a, b, c]
Zimbabwe	14,900	22,000	480,000[b]

[a] Brine, with a lithium content of approximately 0.15 wt.%;

[b] Spodumene, with a Li_2O content higher than 1.0 wt.%, up to 8.0 wt.%;

[c] Lepidolite, with a Li_2O content ranging from 0.2 wt.% to 0.53 wt.%, averaging 0.4 wt.%

Materials and Methods

System Boundaries and Functional Unit

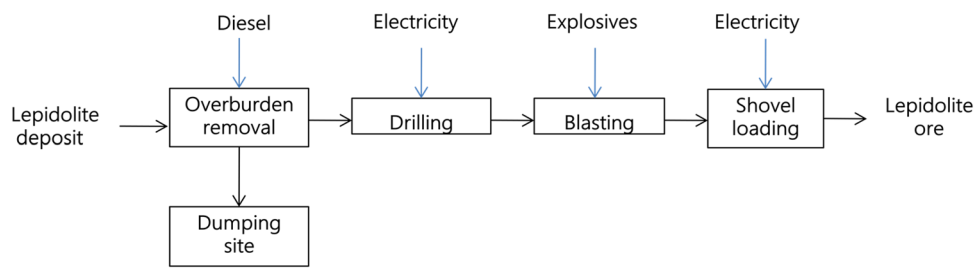
The production process of battery-grade lithium carbonate from lepidolite mainly includes three stages: lepidolite ore mining, beneficiation, and lithium carbonate production. The

corresponding system boundaries and scope are shown in Fig. 1.

The mining method for lepidolite ore is open-pit mining, which follows the principle of "simultaneous stripping and mining, with stripping taking precedence [30]." Before mining begins, the surface overburden is stripped away, and the process proceeds from top to bottom. The stripping method combines mechanical and manual stripping. First, the topsoil is removed to create a working platform, and then the mining proceeds in steps downward. For fully weathered lithium ore, non-blasting operations are used, with excavators directly digging the ore. For weakly weathered or unweathered lithium ore, the mining and stripping process includes drilling, blasting, loading, and transportation. Both ore and waste soil transportation within the mining area are carried out using dump trucks (Figs. 2, 3).

After mining, the lepidolite ore is transported to the beneficiation plant, where it is crushed to a particle size of approximately 5 to 12 mm using a jaw crusher. The crushed ore is then screened to a size below 8 mm and subjected to wet ball milling with water and steel balls. After ball milling, the ore particles with a size less than 0.2 mm enter the magnetic separation process, with the discharge being iron slag. The ore particles after magnetic separation are conveyed to the cyclone screening and flotation processes, resulting in a pulp with a concentration of about 30%. After dewatering, a lepidolite concentrate with a moisture content of 15% is obtained, and the Li_2O content in the lepidolite concentrate

Fig.1 Lepidolite mining process



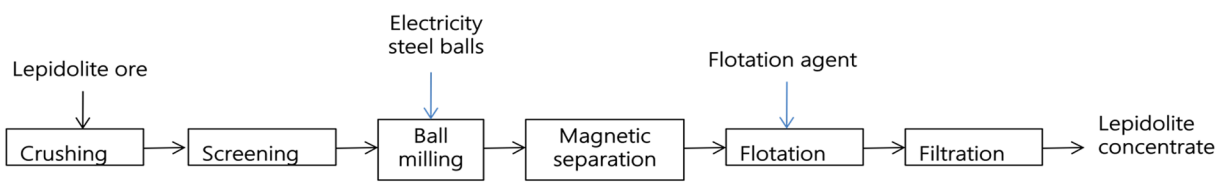


Fig. 2 Lepidolite beneficiation process

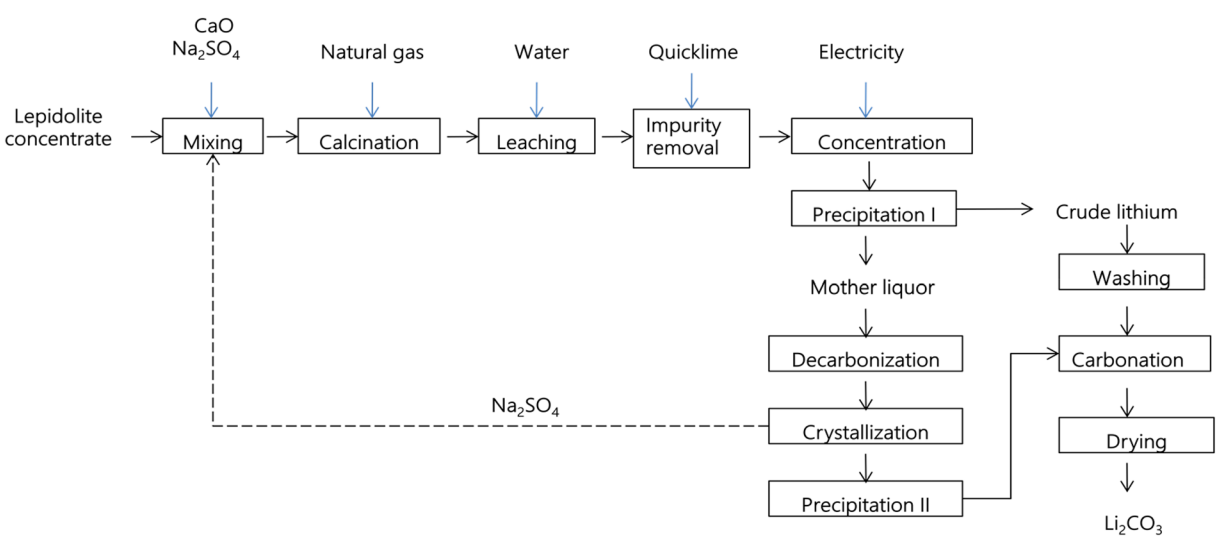


Fig. 3 The process of producing lithium carbonate from lepidolite using the sulfate method

is approximately 2.9% [31]. Some lepidolite beneficiation processes also include the selection of tantalum and niobium concentrates [32].

There are various processes for producing lithium carbonate from lepidolite concentrate, including acid methods, alkali methods, salt roasting, and pressure cooking, among others. In China, the sulfate roasting method is commonly used [21, 33]. In this method, lepidolite concentrate is mixed with sodium sulfate, calcium sulfate, calcium oxide, and other materials, and then roasted in a rotary kiln at a controlled temperature of 800 to 1000 °C. At this temperature, the main components of the raw materials are in a molten state, while minerals with higher melting points remain in solid form. The molten lepidolite undergoes an ion exchange reaction with the sulfates. Calcium oxide is primarily used to remove fluorine from lepidolite, producing calcium fluoride, which is present in the roasting residue. A small amount of fluorine is released as hydrogen fluoride gas with the roasting exhaust gases. The fuel for roasting is natural gas. The main chemical reactions occurring during the roasting process are shown as formula (1) to (3).

$$\begin{aligned} &2(Li,K)_2O \cdot 2Al_2O_3 \cdot 6SiO_2 \cdot 2H(OH,F) \\ &\quad + 2Na_2SO_4 --- 2Li_2SO_4 + 2SiO_2 \\ &\quad + 2K_2SiO_3 + 4NaAlSi_2O_6 + 4Al_2O_3 \cdot SiO_2 \\ &\quad + 2HF \uparrow + 3H_2O \uparrow \end{aligned} \tag{1}$$

$$\begin{aligned} &2(Li,K)_2O \cdot 2Al_2O_3 \cdot 6SiO_2 \cdot 2H(OH,F) \\ &\quad + 2CaSO_4 - 2Li_2SO_4 + 2SiO_2 + 2K_2SiO_3 \\ &\quad + 2Ca\left(AlSi_2O_6\right)_2 + 4Al_2O_3 \cdot SiO_2 + 2HF \uparrow + 3H_2O \uparrow \end{aligned} \tag{2}$$

$$CaO + HF --- CaF_2 + H_2O \uparrow \tag{3}$$

The roasting products are leached with water, and the generated lithium sulfate enters the aqueous solution, forming lithium-bearing mother liquor. After impurity removal, calcium removal, concentration, and ion exchange, the solution contains high-concentration, high-purity lithium ions. A high-purity sodium carbonate solution is then added to the solution to react with the lithium ions, producing lithium carbonate. Since the solubility of Li_2CO_3 decreases with increasing temperature, the solution temperature is continuously increased through indirect heating by steam coils in the lithium carbonate precipitation tank to maximize the precipitation of Li_2CO_3. The reaction mechanism is as formula (4):

$$Li_2SO_4 + Na_2CO_3 = Li_2CO_3 + Na_2SO_4 \tag{4}$$

After precipitation from the mother liquor, centrifugation is used to obtain crude lithium carbonate. The crude lithium carbonate is then washed and dried to produce battery-grade lithium carbonate. The mother liquor after precipitation undergoes decarbonization by adding concentrated sulfuric acid, evaporation and crystallization, freezing, and secondary precipitation processes to recover sodium sulfate, potassium sulfate, and rubidium-cesium mother liquor. Sodium sulfate and potassium sulfate, apart from being partially reused, are sold externally along with the rubidium-cesium mother liquor.

Therefore, the lithium extraction process from lepidolite differs from the previously reported process of extracting lithium from spodumene. Lithium extraction from spodumene generally employs the sulfuric acid method [26], while lithium extraction from lepidolite typically uses the sulfate method [21]. Since spodumene usually does not contain fluorine, there is generally no need to add calcium oxide for fluorine removal. In contrast, fluorine removal is necessary during the lepidolite lithium extraction process [34]. Additionally, during the lepidolite lithium extraction process, sodium sulfate, potassium sulfate, and rubidium-cesium mother liquor can also be recovered simultaneously to enhance the overall economic benefits of the products [35].

Data Collection

The input data related to the mining, beneficiation, and production of lithium carbonate from lepidolite is derived from field surveys, while the background data are sourced from the Ecoinvent 3.8 database [36]. The lepidolite mine, beneficiation plant, and lithium carbonate production facility are all located in Yichun City, Jiangxi Province, China, with transportation distances not exceeding 50 km. Therefore, the transportation of raw materials is not considered in the modeling. Other reagents and chemicals used in the lithium extraction process from lepidolite are purchased locally, and their transportation distances are not considered due to the relatively small quantities used. The natural gas used for roasting is supplied through the West–East Gas Pipeline, originating from Xinjiang. It is transported to the project site via the natural gas pipeline network, which is funded by the government as part of infrastructure construction. Due to the difficulty in obtaining data, the natural gas pipeline network is not considered in the modeling. Electricity used in each stage is purchased from the Central China Power Grid (CCPG), and the life cycle modeling employs background data from the CCPG in 2021. Factory buildings and fixed assets related to lithium extraction are not considered due to the difficulty in obtaining data.

The lepidolite ore deposit is located in Yichun City, Jiangxi Province, China, with a Li_2O content of 0.35 wt.%, which is lower than the average Li_2O content of lepidolite ores (0.4 wt.%) [21]. Mining is conducted via the open-pit

method. The beneficiation process employs a series of operations including crushing, screening, ball milling, magnetic separation, and flotation, which collectively represent a standard industrial practice. Finally, Li_2CO_3 production is carried out using the sulfate method, a well-established process for lithium extraction from lepidolite. Given the very similar technologies used to extract lithium from lepidolite, the life cycle inventory (LCI) data employed in this study is broadly representative of the industry.

Environmental Impact Assessment Method

The LCA model was developed using OpenLCA software (version 2.1.1), with environmental impacts evaluated via the ReCiPe 2016 method [37]. This method converts life cycle inventory data into 18 midpoint indicators and three endpoint indicators—human health, ecosystem quality, and resource availability. To ensure comparability with the growing body of LCA literature on lithium-ion batteries and electric vehicles [38–41], this study employed the ReCiPe 2016 Midpoint (H) method to characterize the 18 midpoint impact categories and the ReCiPe 2016 Endpoint (H) method to calculate the 3 endpoint impacts.

Data Quality and Uncertainty Quantification

The data quality was assessed according to criteria including reliability, completeness, and temporal, geographical, and technological correlations [42]. To evaluate the influence of data variability on the model outcomes, a quantitative sensitivity analysis was conducted. The data were classified into two quality tiers: high-quality data, obtained mainly from field investigations, and low-quality data, sourced from the literature, the Ecoinvent database, or other proxy sources. Prior distributions of 10% and 30% were assigned to the high- and low-quality data tiers, respectively. A Monte Carlo simulation with 5,000 runs was then performed by randomly sampling input parameters from their corresponding probability distributions. This methodology is well-established in LCA for uncertainty analysis [43].

Sensitivity Analysis

To identify key influential parameters, this study conducted a sensitivity analysis. The Li_2O content in lepidolite ore is a critical factor affecting lithium extraction. Generally, a higher Li_2O content is preferable, while ore with excessively low Li_2O levels lacks commercial viability. As the Li_2O content decreases, mining requires more diesel consumption, and the beneficiation and metallurgical stages demand increased energy and chemical inputs, resulting in higher pollutant emissions. Therefore, it is essential to analyze how variations in Li_2O content across different lepidolite ores influence various environmental impact categories of lithium extraction.

China's electricity mix remains heavily dominated by thermal power, which is a major reason for the currently high carbon emissions associated with lithium-ion batteries and electric vehicles. With the vigorous development of renewable energy in China, particularly under the national carbon neutrality target, the share of coal power is expected to decline gradually, leading to a greener electricity structure. This transition will significantly enhance the environmental performance of lithium extraction from lepidolite. To quantitatively assess the impact of China's evolving electricity mix on lepidolite-based lithium extraction, this study evaluates the environmental implications using the projected electricity structures of 2025, 2030, and 2050.

Economic Analysis

A techno-economic evaluation of lithium extraction from lepidolite was conducted using the Techno-Economic Cost Model (TCM) [44]. The TCM methodology functions as a decision-making tool, enabling economic comparisons among different technological routes by focusing solely on operating costs and benefits to identify the most advantageous option, without taking into account capital construction expenses [45]. In this model, operating costs primarily arise from the consumption of raw materials, chemicals, and energy. Revenue is estimated based on benchmark prices for the main products and by-products, with all pricing data sourced from surveys carried out on Chinese online B2B platforms [46–49]. Unit prices from December 2024 were used as reference values (Table 2).

Results and Discussion

Life Cycle Inventory

The functional unit for the mining phase, the beneficiation phase, and the lithium carbonate production phase is defined as the production of 1.0 kg of battery-grade Li_2CO_3. The cradle-to-gate life cycle inventories for these three phases are presented in Table 3, respectively.

The mining method for lepidolite is open-pit mining. The mining equipment used includes drilling rigs, rock drills, breaker hammers, excavators, loaders, bulldozers, and dump trucks. In addition to the raw material inputs listed in the table, based on the annual mining of 3 million tons of lepidolite, 10 drill rods, 700 drill bits, and 30,000 detonators are consumed each year. However, since these quantities are relatively small when allocated to functional units, their consumption is considered negligible. The domestic water for the mine workers is sourced from the municipal

Table 2 Unit price for input and output calculations. (Functional unit: 1 kg Li_2CO_3)

Inputs	Unit	Base price	Low price	High price
Explosives	\$/kg	0.45	0.36	0.56
Steel balls	\$/kg	0.45	0.42	0.57
Organic amine flotation reagent	\$/kg	1.40	1.12	1.69
Sodium hexametaphosphate	\$/kg	1.26	0.98	1.54
PAM	\$/kg	1.33	0.70	1.69
Polyaluminum chloride	\$/kg	0.22	0.13	0.25
Soda ash	\$/kg	0.17	0.12	0.20
Quicklime	\$/kg	0.08	0.07	0.11
NaOH	\$/kg	0.14	0.10	0.15
Sulfuric acid	\$/kg	0.04	0.04	0.09
Gypsum	\$/kg	0.01	0.01	0.03
CO_2	\$/kg	0.28	0.22	0.31
Diesel	\$/MJ	0.22	0.22	0.24
Electricity	\$/kWh	0.08	0.06	0.15
Natural gas	$\$/m^3$	0.59	0.59	0.62
Water	\$/kg	0.0002	0.0002	0.0003
Outputs				
Li_2CO_3	\$/kg	11.24	5.62	80.06
Tantalum	\$/kg	287.43	244.45	327.74
Niobium	\$/kg	60.23	56.12	65.43
Na_2SO_4	\$/kg	0.08	0.06	0.10
K_2SO_4	\$/kg	1.69	1.40	1.97
Rubidium	\$/kg	702.25	280.90	842.70
Cesium	\$/kg	126.40	98.31	224.72
Wastewater treatment	\$/kg	0.0006	0.0006	0.0006
Solidwaste treatment	\$/kg	0.03	0.03	0.03

water supply network. The production water is sourced from treated mine pit inflow and waste rock dump leachate, which are not included in the raw material inputs.

This is likely the first time we have reported the input–output inventory of the lepidolite mining process. Although lithium extraction from spodumene is more common, data on the mining of spodumene are rarely reported. When tracing the background data for spodumene mining, Stamp et al. [9] used the ecoinvent process "iron ore, 46% Fe, at mine/kg/GLO" from the Ecoinvent V2.2 database as a substitute data source. Clearly, this is a stopgap measure for LCA modeling when data are lacking. However, the mining process of iron ore is different from that of lepidolite. Establishing an input–output inventory for lepidolite mining is meaningful for improving the life cycle assessment of lithium-ion batteries.

The beneficiation process generally adopts a wet method, with the aim of increasing the lithium oxide content in the concentrate. The raw material inputs in the beneficiation stage mainly include raw lepidolite ore, steel balls, and flotation reagents. Steel balls are primarily used in the ball milling process to grind the lepidolite ore into fine particles. To improve the surface properties of the ore particles, flotation reagents are typically added during the mineral processing. The flotation reagents are formulated with a 1:1 ratio of organic amine and sodium hexametaphosphate. The organic amine flotation reagent has the molecular formula $CH_3(CH_2)_{11}NH_2$ and is produced by reacting dodecanoic acid with ammonia to form dodecanenitrile, which is then catalytically hydrogenated under pressure to form enamine and further reduced. Since the data for this chemical are missing in Ecoinvent 3.8, a commonly used organic amine in detergents is used as a substitute. Sodium hexametaphosphate, with the molecular formula $(NaPO_3)_6$, can form soluble complexes with metal ions such as calcium and magnesium. Since the data for this chemical are also missing in Ecoinvent 3.8, sodium phosphate is used as a substitute. In addition, the laboratory uses small amounts of hydrofluoric acid, sulfuric acid, and sodium hydroxide. The output of the mineral processing stage is lepidolite concentrate. The pollutants emitted include dust into the atmosphere, and the solid waste mainly consists of tailings and iron slag.

During the mineral processing stage of spodumene, Stamp et al. [9] used the ecoinvent process "manganese concentrate, at beneficiation/kg/GLO" from Ecoinvent V2.2 as a substitute data source. The authors did not explain why

Table 3 Life cycle inventory of Li_2CO_3 produced from lepidolite ore. (Functional unit: 1 kg Li_2CO_3)

	Input		Output	
Mining stage	Explosives/kg	0.05	Lepidolite ore/kg	189.00
	Diesel/MJ	60.94		
	Electricity/kWh	0.17		
Beneefication stage	Lepidolite ore/kg	189.00	Lepidolite concentrate /kg	18.00
	Steel balls/kg	1.39E-01	Dust/kg	3.65E-04
	Organic amine flotation reagent /kg	1.19E-01	Tailings mud /kg	4.41
	Sodium hexametaphosphate /kg	9.90E-02	Iron slag /kg	0.40
	PAM/kg	2.70E-03	Tantalum /kg	3.06E-03
	Polyaluminum chloride /kg	1.19E-01	Niobium /kg	2.52E-03
	Electricity/kWh	75.60		
	Water/kg	27.36		
Li_2CO_3 production stage	Lepidolite concentrate/kg	18.00	Li_2CO_3/kg	1.00
	Soda ash/kg	2.50	COD/kg	3.72E-04
	Quicklime /kg	1.50	SS/kg	7.45E-05
	NaOH/kg	0.01	Fluoride/kg	1.25E-05
	Sulfuric acid /kg	1.50	NH_3-N/kg	3.73E-05
	Gypsum /kg	2.00	TP/kg	3.73E-06
	CO_2/kg	0.50	Thallium/kg	2.24E-08
	Electricity/kWh	4.32	Sulfuric acid mist/kg	4.65E-05
	Natural gas/m^3	2.33	SO_2/kg	4.15E-03
	Water/kg	0.04	NOx/kg	3.48E-03
			TSP/kg	2.60E-03
			Na_2SO_4/kg	0.50
			K_2SO_4/kg	0.50
			Rubidium/kg	0.02
			Cesium/kg	5.40E-03

they used the manganese concentrate processing to replace spodumene. In fact, they used the iron ore mining process as an approximation. Clearly, using the actual input–output data for spodumene processing is the best way to reduce data uncertainty. Kelly et al. [27] used actual production data for the spodumene concentrate processing, but only included diesel and not electricity, which is consistent with the actual situation of mineral processing in Australia. However, China has a well-developed power grid, and electricity is generally used in the lepidolite beneficiation stage in China. Therefore, the spodumene processing data from Australia do not match the actual situation of lepidolite beneficiation in China.

In the stage of producing lithium carbonate from lepidolite concentrate, the sulfate method is generally adopted. Sodium sulfate and quicklime are added to the lepidolite concentrate and mixed evenly before being fed into a rotary kiln for high-temperature calcination. Lithium in lepidolite reacts with sodium sulfate at high temperatures to produce lithium sulfate. Sodium sulfate is required for the initial production and is recycled for subsequent use. Quicklime is primarily used to remove fluorine elements from lepidolite concentrate. The generated lithium sulfate enters the aqueous solution through the leaching process. Quicklime is added to adjust the pH value of the solution to remove calcium and iron ions, and sodium carbonate is added to completely precipitate calcium ions. Additionally, a small amount of chelating resin is used to remove calcium and magnesium ions. When the ion exchange resin becomes saturated, it is regenerated with sulfuric acid and sodium hydroxide for continued use. Since the amount of ion exchange resin used is relatively small, it is not accounted for in this process. In the lithium-containing solution, high-purity Na_2CO_3 is added to react with lithium ions to produce Li_2CO_3. The crude Li_2CO_3 and primary lithium precipitation mother liquor are obtained through centrifugation. The crude Li_2CO_3 is washed with pure water, pulped, and reacted with carbon dioxide, followed by centrifugation and drying to obtain battery-grade high-purity Li_2CO_3. In the primary lithium precipitation mother liquor, concentrated H_2SO_4 is added for decarbonization, and NaOH is used to adjust the pH value. Na_2SO_4 is recovered through concentration and evaporative crystallization. Part of the Na_2SO_4 is sold externally, while the remainder is reused in the calcination process. Natural gas is generally used for the calcination of

lepidolite concentrate. The pollutants emitted include water pollutants such as COD, SS, fluoride, ammonia nitrogen, TP, thallium, etc., and gaseous pollutants such as sulfuric acid mist, SO_2, NO_x, TSP, etc. The main solid waste generated is lithium slag.

Environmental Impacts

The environmental impacts of Li_2CO_3 production from lepidolite were assessed using the ReCiPe 2016 Midpoint (H) and Endpoint (H) method, with results shown in Table 4. To facilitate comparison with the environmental impacts of Li_2CO_3 production from spodumene, a modeling and environmental impact calculation were conducted based on the production data of Tianqi Lithium (using Australian spodumene concentrate, see as Table S1 and Table S2) [27]. The comparative results are also presented in Table 4.

The results indicate that lepidolite exhibits poorer environmental performance than spodumene across all 18 midpoint and 3 endpoint impact categories. The primary reasons for this are as follows:

(1) Lower ore grade of lepidolite, which necessitates higher energy consumption during mining. In this study, the lithium oxide (Li_2O) content in lepidolite is only 0.35 wt.%, compared to 1.9 wt.% in spodumene. Producing one functional unit of lithium carbonate requires mining 189 kg of lepidolite ore, whereas only 33 kg of spodumene ore is needed. Consequently, the energy inputs (primarily diesel) for lepidolite mining is nearly double that for spodumene.

(2) Higher impurity content of lepidolite, requiring more chemical reagents during purification. In addition to neutralizing acidic wastewater, the lithium extraction process from lepidolite demands extra quicklime for fluorine removal, with quicklime consumption more than double that of the spodumene process. Lepidolite beneficiation also consumes more flotation reagents, which are significant contributors to ecosystem toxicity [50]. Furthermore, the process generates more flotation wastewater than spodumene, and treating this effluent requires additional chemicals such as polyacrylamide (PAM), a notable contributor to greenhouse gas emissions [51].

Table 4 Comparison of environmental impact indicators for Li_2CO_3 from lepidolite (LL) and spodumene (LS) based on the ReCiPe 2016 Midpoint (H) and Endpoint (H) method

	Impact category	Abbreviation	Reference unit	LL*	LS**	LL/LS
Midpoint indicators	Fine particulate matter formation	FPMF	kg $PM_{2.5}$ eq	0.12	0.02	6.56
	Fossil resource scarcity	FRS	kg oil eq	13.63	4.53	3.01
	Freshwater ecotoxicity	FEC	kg 1,4-DCB	2.14	1.15	1.87
	Freshwater eutrophication	FEU	kg P eq	0.01	0.00	6.36
	Global warming	GW	kg CO_2 eq	66.40	5.78	11.49
	Human carcinogenic toxicity	HCT	kg 1,4-DCB	4.04	0.56	7.18
	Human non-carcinogenic toxicity	HNT	kg 1,4-DCB	74.71	35.68	2.09
	Ionizing radiation	IR	kBq Co-60 eq	3.56	0.18	20.13
	Land use	LU	m^2a crop eq	1.27	0.33	3.88
	Marine ecotoxicity	MEC	kg 1,4-DCB	2.81	1.51	1.86
	Marine eutrophication	MEU	kg N eq	1.82E-03	1.12E-04	16.28
	Mineral resource scarcity	MRS	kg Cu eq	63.79	0.09	711.22
	Ozone formation, Human health	OF(HH)	kg NOx eq	0.23	0.04	5.60
	Ozone formation, Terrestrial ecosystems	OF(TE)	kg NOx eq	0.23	0.04	5.56
	Stratospheric ozone depletion	SOD	kg CFC11 eq	2.11E-05	3.10E-06	6.82
	Terrestrial acidification	TA	kg SO_2 eq	0.28	0.04	6.55
	Terrestrial ecotoxicity	TEC	kg 1,4-DCB	232.70	177.38	1.31
	Water consumption	WC	m^3	0.29	0.06	4.59
Endpoint indicators	Human health		DALY	1.70E-04	2.73E-05	6.23
	Ecosystems		species.yr	3.03E-07	3.86E-08	7.84
	Resources		USD2013	16.83	7.79E-01	21.61

LL*: lithium extraction from lepidolite. LS**: lithium extraction from spodumene

(3) Divergent lithium extraction processes. The spodumene-based process typically involves high-temperature roasting (1200 °C) followed by H_2SO_4 leaching [52], whereas the lepidolite-based process generally employs low-temperature sulfate roasting (700 °C) with subsequent water leaching [53]. Accordingly, the lepidolite process is more energy-efficient, consuming approximately 85% of the energy required for spodumene processing. While this energy advantage modestly improves its environmental profile, it is insufficient to offset the impacts from mining and purification.

In recent years, China has been actively expanding its renewable energy capacity, with a steadily increasing share of green electricity in its power mix [54]. The adoption of photovoltaic panels in lepidolite mining areas and the use of electrified mining equipment could reduce the endpoint environmental impacts of lepidolite-based lithium extraction—specifically in Human Health, Ecosystems, and Resources—by 17%, 15%, and 5%, respectively. Moreover, employing renewable energy throughout the entire lithium extraction process could achieve further reductions of 89%, 92%, and 12% in these respective endpoint impact categories.

Contribution Analysis

Contribution of Each Stage

The production of Li_2CO_3 from lepidolite ore involves three stages: mining, beneficiation, and Li_2CO_3 production. The environmental impact of the third stage is normalized to 100%, and a comparison of the environmental impacts across the three stages is presented in Fig. 4. Among the 18 midpoint environmental impact categories, mining contributes most significantly to LU, accounting for 52.17%, followed by HNT at 40.45%. Mining also accounts for 27.94%, 28.15%, and 27.86% of OF(HH), OF(TE), and SOD, respectively. For the remaining 13 impact categories, the contribution from mining does not exceed 20%. In terms of the 3 endpoint indicators—Human Health, Ecosystems, and Resources—mining contributes 17.37%, 15.32%, and 5.27%, respectively. Previous studies on lithium extraction from ore [9, 26] have often overlooked the environmental impact of the mining stage due to a lack of data, which may lead to an underestimation of the overall life cycle environmental impact. Therefore, the environmental impact of the lepidolite ore mining stage should not be neglected. The beneficiation stage accounts for 76.83%, 81.06%, and 8.09% of the endpoint environmental impacts in human health, ecosystem quality, and resource availability, respectively. At the midpoint level, the impact categories most significantly influenced by the beneficiation stage are IR, WC, and MEU, contributing 90.29%, 89.33%, and 88.20%, respectively. In contrast, the Li_2CO_3 production stage contributes 5.80%, 3.62%, and 86.64% to the same endpoint impact categories. The midpoint categories most affected by this stage are MRS and TEC, accounting for 100% and 57.16%, respectively. The rationale for attributing MRS almost entirely to the third stage stems from its calculation methodology, which employs the Surplus Ore Potential (SOP) as the characterization factor. The SOP principle states that progressive exploitation depletes higher-grade ores; thus, future extraction of an equivalent resource unit will require processing a greater volume of lower-grade ore [36]. Given that mining and beneficiation are merely physical processes, and the third stage (Li_2CO_3 production) is where chemical transformation occurs, the MRS is allocated accordingly.

Contribution of Materials and Energy

The contribution of materials and energy inputs to various categories of environmental impact is shown in Fig. 5. Electricity serves as the primary contributor to FPMF, FRS, FEU, GW, IR, OF(HH), OF(TE), SOD, and TA, accounting for over 60% of the impact in each category. It exhibits the highest influence in the IR, GW, and FRS categories, with

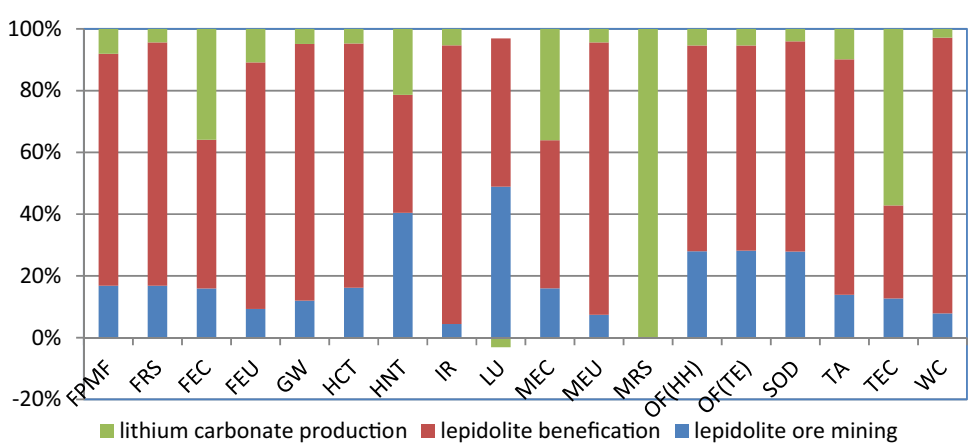


Fig. 4 Comparison of environmental impacts of lithium extraction from lepidolite at the stages of ore mining, beneficiation, and lithium carbonate production, with the impact of the third stage as 100%

contribution values of 92.74%, 84.49%, and 79.23%, respectively. Additionally, electricity is the dominant contributor to WC, HCT, MEC, and FEC, with shares of 49.98%, 44.19%, 40.03%, and 39.70%. It also ranks as the second most significant contributor to HNT and TEC, representing 34.87% and 17.11%, respectively. China's coal-dominated energy structure is a significant contributor to global warming and other environmental impacts [55]. However, with the increasing integration of renewable energy, its electricity supply is expected to become progressively greener [56]. Diesel is the leading contributor to LU and HNT, with contributions of 50.88% and 38.87%. It is also the second most important contributor after electricity in categories including FPMF, FRS, GW, IR, OF(HH), OF(TE), SOD, and TA. Notably, its contribution to OF(TE), OF(HH), and SOD reaches 28.00%, 27.80%, and 17.83%, respectively. Furthermore, diesel is the third largest contributor to HCT and TEC, at 15.33% and 11.61%. The high combustion temperature and propensity for incomplete combustion in diesel engines are key drivers behind the severity of particulate matter pollution and its toxicity [57]. Sulfuric acid is identified as the primary contributor to TEC, accounting for 57.09%. It also ranks as the second largest contributor to both MEC and MRS, with shares of 36.48% and 26.04% respectively. Additionally, it represents the third most significant contributor to HNT, constituting 21.33% of the impact in this category. China, the world's top sulfuric acid producer, generates acid primarily from sulfur (44.1%), smelting operations (35.3%), and ore processing (17.9%) in 2022 [58]. However, SO_2 emissions from production flue gas persist as a significant source of air pollution across all methods [59]. Steel balls consumed in the ball mill are a significant source of environmental impact, contributing primarily to MRS (30.64%) and HCT (29.26%), with an additional 9.01% contribution to TEC. This high material and energy footprint is a characteristic feature of the steel industry's environmental profile [60]. Organic flotation reagents represent the largest share of the MEU impact (48.90%), alongside substantial contributions to LU (39.08%) and WC (22.70%). The prevalence of synthetic chemicals in these reagents raises concerns about environmental and safety risks, notably the persistence and toxicity of the flotation wastewater, which is notoriously difficult to treat [61].

Uncertainty and Monte Carlo Analysis

The foreground data for this study were obtained from annual production records provided by suppliers of lepidolite ore, concentrate, and Li_2CO_3. This approach ensures the representativeness and reliability of the key data inputs for the assessment. Several methodological assumptions were necessarily incorporated, including the local procurement of all raw materials, the use of background datasets from the Ecoinvent database, and the application of proxy data for organic amines and sodium hexametaphosphate. These choices intrinsically introduce a degree of uncertainty into the model. Nonetheless, the LCA results reveal that the impact of these assumptions on the overall findings is marginal.

As summarized in Table 5, the Monte Carlo analysis indicates that all environmental impact categories exhibit coefficients of variation (CV) below 17% at a 90% confidence level. Thus, the combined evidence from the uncertainty and sensitivity analyses substantiates the robustness of the study's conclusions with respect to variations in data input.

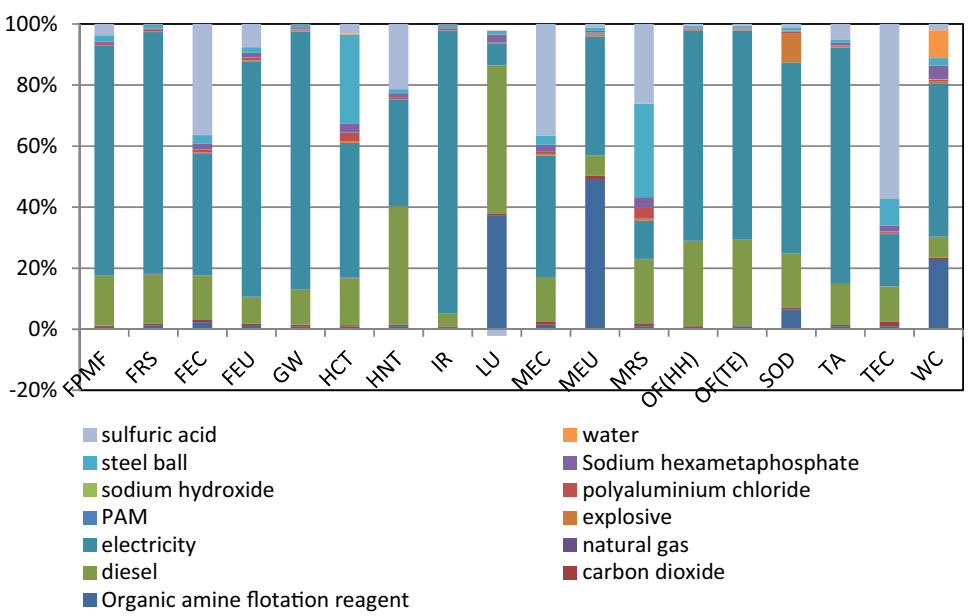


Fig. 5 Contribution of materials and energy inputs to different environmental impact categories for Li_2CO_3 production from lepidolite

Sensitivity Analysis

Variation of Li_2O Content in Lepidolite Ore

The Li_2O content in raw lepidolite ore typically ranges from 0.1% to 0.8% [62], although lower-grade ores may not be economically viable for extraction. This study assumes a consistent beneficiation recovery rate and evaluates the environmental impacts of producing Li_2CO_3 from ores with varying Li_2O contents. The corresponding range of results is shown in Fig. 6. When lower-grade lepidolite ore (with 0.1 wt.% Li_2O) is used as the raw material, the environmental impacts increase significantly—particularly for impact categories such as LU, HNT, OF(HH), OF(TE), and SOD—with a maximum increase of 92%. Conversely, when higher-grade ore (with 0.8 wt.% Li_2O) is used, the environmental impacts are markedly reduced in the same categories, with reductions of up to 34%.

Variation of Electricity Mix

Currently, China's electricity sector is dominated by coal power, which represents a major source of air pollution [63]. The electricity data used in this study were obtained from the Ecoinvent 3.8 database, based on China's 2021 mixed electricity dataset, with coal-fired power constituting 80.2% of the mix. In alignment with China's energy development strategy, the electricity mix is set to be optimized through the vigorous expansion of renewable energy sources, including photovoltaic and wind power [64]. As a result, the share of coal power is projected to decline to 48.8%, 26%, and 5.7% by 2025, 2030, and 2050, respectively [65] (detailed electricity mix data are provided in the online supplementary material). Using the 2021 electricity structure as the baseline, Fig. 7 illustrates how changes in the electricity mix across different years influence the environmental footprint of lithium extraction from lepidolite. The results demonstrate that the ongoing decarbonization of the power supply will significantly reduce the environmental burden associated with lepidolite-based lithium extraction. The environmental impact categories with the most pronounced reductions are IR, GW, and FRS. By 2050, their corresponding values are projected to drop to only 13%, 21%, and 26% of the baseline levels, respectively.

Cost Analysis of Lithium Extraction from Lepidolite

In lepidolite ore, lithium is typically associated with minerals such as tantalum, niobium, rubidium, and cesium. Lithium extraction from lepidolite generally follows two routes in China: with or without resource recovery. Figure 8 presents the cost, revenue, and profit analysis of the two lithium extraction routes under base, low, and high price. It can be observed that at the base price, when resource recovery is not implemented, the revenue from lithium extraction only covers half of the cost, resulting in a negative profit and poor economic feasibility. In contrast, with resource recovery, revenue exceeds cost, yielding a positive profit and significantly improving economic viability. From 2022 to 2024, international lithium carbonate prices experienced considerable

Table 5 Table results of the Monte Carlo simulations for the lithium extraction from lepidolite

Impact category	mean	SD	CV	90%CI
FPMF	0.12	0.02	12.20%	[0.099;0.148]
FRS	13.66	1.72	12.61%	[10.918;16.583]
FEC	2.15	0.19	8.81%	[1.841;2.460]
FEU	0.01	0.00	11.88%	[0.010;0.016]
GW	66.54	8.50	12.77%	[52.876;80.975]
HCT	4.05	0.46	11.25%	[3.313;4.809]
HNT	74.87	7.90	10.55%	[62.453;88.289]
IR	3.57	0.47	13.16%	[2.814;4.369]
LU	1.27	0.21	16.21%	[0.943;1.617]
MEC	2.81	0.25	8.83%	[2.413;3.222]
MEU	1.82E-03	2.95E-04	16.22%	[1.350E-3;2.335E-3]
MRS	63.79	0.01	0.02%	[63.774;63.812]
OF(HH)	0.23	0.03	12.17%	[0.185;0.277]
OF(TE)	0.23	0.03	12.12%	[0.187;0.279]
SOD	0.00	0.00	12.12%	[1.704E-5;2.555E-5]
TA	0.28	0.03	11.91%	[0.224;0.333]
TEC	233.00	18.13	7.78%	[2.037E2;2.634E2]
WC	0.29	0.04	12.63%	[0.234;0.355]

SD standard deviation, *CV* coefficient of variation, *CI* confidence interval

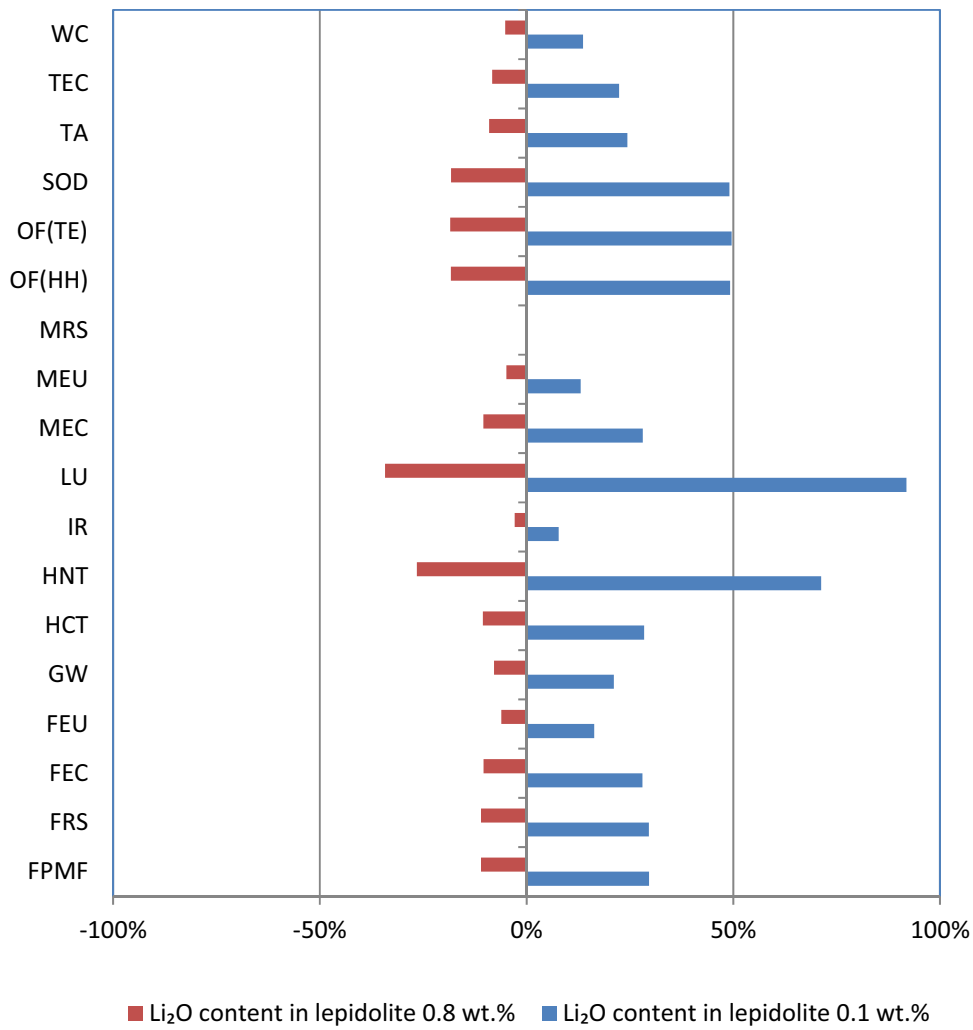


Fig. 6 Sensitivity analysis of the environmental impacts on Li_2O content in lepidolite ore

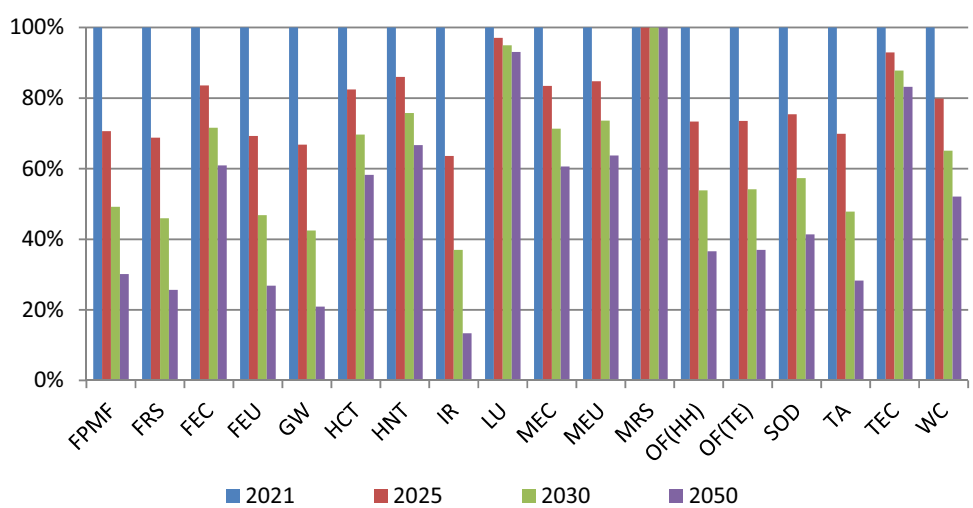


Fig. 7 Sensitivity analysis of the environmental impacts on electricity mix in China at 2021, 2025, 2030, and 2050

fluctuations. At the lowest price point of just US$5.62 per kg, revenues from both extraction routes were insufficient to cover costs, regardless of resource recovery, leading to negative profits. Particularly in the absence of resource recovery, revenue accounted for only 30% of the cost. At the peak price, however, revenues from lepidolite-based lithium

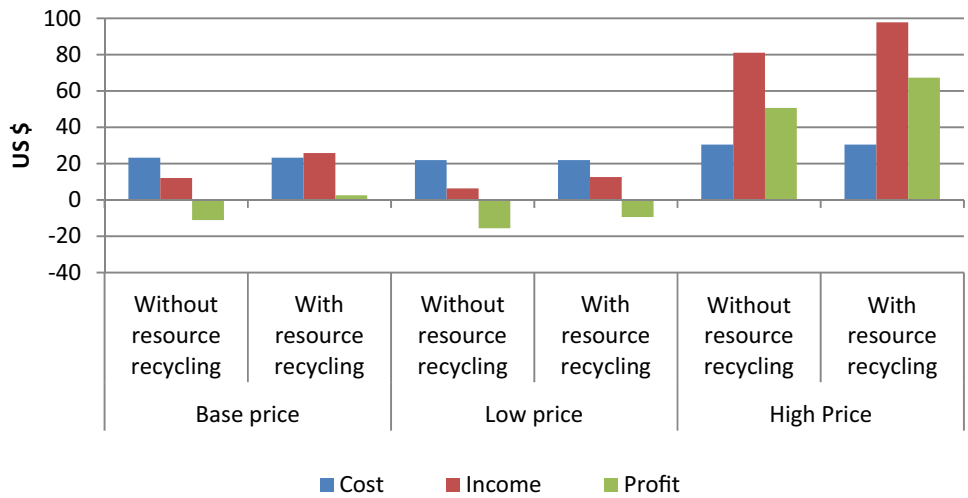


Fig. 8 Comparison of the economic costs, incomes and profits without/with resource recycling at base, low, and high price

extraction exceeded costs under both routes. Notably, when resource recovery was incorporated, the profit reached 2.2 times the cost, demonstrating strong economic performance. These findings indicate that the price of lithium carbonate exerts a substantial influence on the economic feasibility of lithium extraction from lepidolite. Moreover, comprehensive resource recovery plays a critical role in significantly enhancing its economic attractiveness [21].

Limitations and Future Work

This study presents a comprehensive analysis of the environmental impacts and techno-economic performance of lithium extraction from lepidolite. Nevertheless, the study has several limitations.

First, it does not account for potential variations in mining methods arising from differences in the lithium oxide content and geological settings of lepidolite ores. Variations in impurity profiles among ores, which influence beneficiation and smelting processes as well as material inputs, were also not considered. Furthermore, the analysis did not incorporate potential transportation costs associated with processing lepidolite concentrates in different regions within the country. These aspects represent constraints in the current work. Future studies should therefore expand the sample set and perform more detailed LCA that explicitly integrate these factors to quantify their influence. Although resource recovery during beneficiation and smelting—which can enhance environmental performance—was included in the LCA model, the study did not allocate environmental impacts to the recovered co-products. The application of allocation weights or specific allocation methodologies was omitted, and the selection of an appropriate allocation approach remains a challenge to be addressed in future research. Regarding life cycle impact assessment (LCIA), this study utilized the ReCiPe midpoint and endpoint methods. Employing other mainstream LCIA methodologies would provide a more comprehensive evaluation of the environmental impacts. Finally, the economic analysis was limited to operational costs, revenues, and profits, excluding capital expenditures and labor costs. A more thorough techno-economic analysis is, therefore, warranted to yield a complete assessment.

Conclusion

Lepidolite is a crucial lithium resource with strategic importance for the supply chain of LiBs raw materials, both currently and in the foreseeable future. This study develops a cradle-to-gate LCA model, with system boundaries spanning from lepidolite mining to the production of battery-grade Li_2CO_3. The principal findings of this study are summarized as follows.

(1) For the first time, a life cycle inventory covering the mining, beneficiation, and Li_2CO_3 production from lepidolite has been established, filling the gap in the life cycle environmental impact assessment of lithium extraction from lepidolite.

(2) A comparative LCA demonstrated that lithium extraction from lepidolite generates greater environmental burdens than spodumene in all 18 midpoint and all three endpoint categories. The endpoint impacts for lepidolite exceeded those for spodumene by a factor of 6.23 for human health, 7.84 for ecosystem quality, and 21.61 for resource depletion. This is owing to the inferior ore grade of lepidolite (lower Li_2O content and higher impurities), leading to disproportionately larger inputs of materials and energy across the entire production chain.

(3) The LCA reveals that the beneficiation stage is the dominant contributor to the Human Health (76.83%) and Ecosystems (81.06%) endpoints, while Li_2CO_3 rpoduction stage contributes most significantly to Resources (86.64%). In contrast, the mining stage exhibits a comparatively minor effect. At the midpoint level, electricity consumption is the primary driver, responsible for over 60% of the impact in categories including FPMF, FRS, and GW. Additionally, diesel is a key contributor to LU and HNT; sulfuric acid to TEC; steel balls to MRS; and organic flotation reagents to MEU.

(4) The environmental footprint of lithium extraction from lepidolite is highly influenced by the ore's Li_2O content and the energy source. A low Li_2O content (0.1%) drastically increases impacts (by up to 90%) in categories like LU and HNT, while a high Li_2O content (0.8%) reduces them (by up to 34%). The greening of China's electricity mix will further mitigate impacts, most notably for IR, GW, and FRS, which are projected to decline to 13%, 21%, and 26% of baseline levels by 2050.

(5) The techno-economic analysis indicates that lithium extraction from lepidolite is currently suboptimal under prevailing lithium carbonate market prices. However, its economic viability can be substantially enhanced by a combination of rising lithium carbonate prices and the implementation of comprehensive resource recovery from the ore.

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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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